SHREEL PATEL

Contact details: +61 448 139 173 | Shreel.p05@gmail.com
LinkedIn: https://www.linkedin.com/in/shreel-s-patel/
PCB Portfolio: https://shreel-pcbportfolio.vercel.app

PROFESSIONAL SUMMARY

Mechatronics Engineering student experienced developing embedded control systems, delivering professional PCB design and documentation, and integrating electro-mechanical systems through work experience and academics. Experienced in analysing system performance, power consumption and design trade-offs to meet strict functional and efficiency requirements. Seeking a hands-on engineering role where reliability, testing and disciplined engineering practice are critical.

PROFESSIONAL EXPERIENCE

Team Arrow Racing Association **November 2025 – Current**
Mechatronics Engineer (Electrical Lead)

- Designed and refined electrical schematics using Altium Designer, supporting power distribution, serial communication integration (CAN) and subsystem integration for a solar powered electric car.
- Selected components defined electrical interfaces and schematic design reviews to improve electrical robustness and system reliability.
- Produced structured engineering documentation, including schematic annotations and test observations to support traceability, design validation and iterative system integration (selected work showcased in the PCB portfolio).
- Designed and optimised LED PCBs to meet specified candela targets while minimising power usage, applying hand calculations and component selection to balance efficiency and performance.

Greg Chappell Cricket Centre **August 2023 – Current**
Part-time Retail Sales Assistant

- Used Microsoft Excel and Microsoft Teams to track warranty cases and stock movement creating an organised backlog for effective issue resolution.
- Worked in a collaborative fast paced work environment, coordinating tasks and priorities to maintain consistent customer service quality during trading hours.

EDUCATION

Queensland University of Technology ***2023 – Current***
Bachelor of Engineering (Honours)
Mechatronics & Computer and Software Systems
Expected Graduation: *November 2026*
Current GPA: 6.217/7
Relevant Coursework: Control and Dynamic Systems, Embedded Systems, Electronics, Signal Analysis, Mechatronics Design.
Awards

- Dean's Commendation for Academic Excellence Award Recipient 2023 & 2025

ENGINEERING PROJECTS

Embedded System for an Electric Vehicle

- Designed and implemented a 3-phase BLDC motor subsystem in C using FreeRTOS, integrating hall effect sensor feedback with closed loop PID control to achieve a <2% steady state RPM error.
- Integrated sensing, system interface and motor control subsystems using RTOS kernel primitives such as semaphores and mutexes to allow for deterministic and safe interthread communication.
- Implemented and validated a rate monotonic scheduling approach to minimize input latency, while ensuring seamless motor and sensor interaction.
- Verified control system stability and timing behaviour through iterative testing and tuning, assessing response under varying conditions.

Honours Thesis – Real to Simulation Transfer of Deformable Plants

- Developed a physics-based deformable plant structure inside of MuJoCo using the mass spring modelling approach for robotic interaction.
- Investigating modelling approaches including the finite element method, rigid body constraints and physics informed neural networks to evaluate their accuracy and computational performance.
- Analysing simulation stability and system behaviour to support large scale vertical farm environments for robotic interaction research.

Autonomous Pick and Pack warehouse Robot

- Designed a robust 3D printed chassis, modelled through Inventor using iterative prototyping to maximise the durability of the robot while minimising the weight.
- Designed and integrated a differential drive system through MATLAB based gear ratio analysis to meet speed and torque requirements.
- Implemented closed loop PID speed control with differential tracked wheels, enabling stable translational and rotational motion while maintaining traction.
- Collaborated in a multidisciplinary team to integrate the electrical mechanical and software subsystems harmoniously to achieve zero functional errors during testing.

ENGINEERING & TECHNICAL SKILLS

Programming: Python, C/C++, C#, MATLAB

Electrical Simulation and Design: Altium Designer, LTspice

Mechanical Design & CAD: AutoCAD, Inventor

Tools: GitHub, Linux, VS Code, Jupyter, FreeRTOS, Altium 365, Microsoft Office

REFERENCE

Upon Request